

FIRECODE: A Computational Chemistry Workflow Driver and Modular Hub

Nicolò Tampellini ¹✉

¹ Massachusetts Institute of Technology, United States of America ✉ Corresponding author

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Summary

Computational chemistry, which seeks to model molecular systems for the extraction of properties of interest, is experiencing a significant renaissance thanks to impressive advances in theoretical models enabling ever-faster simulations of complex processes via density functional theory (DFT), semiempirical methods, and machine-learned interatomic potentials (MLIPs). However, the reduced computational cost of these workflows will only translate into faster and more robust modeling with the help of automation frameworks, where trivial human intervention does not become the new bottleneck of computational pipelines. FIRECODE aims at filling this need by providing a modular framework for generating and processing molecular ensembles.

Statement of need

FIRECODE is a modular workflow driver for computational chemistry written in Python. The code interfaces with a series of fast modern calculators (xTB ([Bannwarth et al., 2019](#)), tblite ([Katbashev et al., 2025](#)), AIMNET2 ([Anstine et al., 2025](#)), UMA ([Wood et al., 2025](#)), usually via ASE ([Bahn & Jacobsen, 2002](#); [Larsen et al., 2017](#))) and exposes a series of workflows and utilities for geometry optimization, frequency analysis, one- and two-ended transition state search, and conformational search. When possible, these routines are implemented in a calculator-independent way, and almost all of these can be arbitrarily chained to one another to define a complex workflow.

The software was born out of necessity over the last five years of research in asymmetric catalysis: FIRECODE was designed to streamline the necessary use of multiple software and routines in the computational modeling of small molecules, in a single centralized hub. The program can be called via command line with a minimal plain-text input file defining the workflow, or used as a Python library in Jupyter notebooks. The modular nature of the software enables rapid addition of new features, facilitating external contributions, while also retaining a small installation footprint for the core library. External programs are easily interfaced via API calls or file-based I/O. The concise input syntax also offers a streamlined definition of workflows and software orchestration with no coding experience required.

State of the field

On top of computational chemistry software, workflow utilities orchestrating sequential calculation steps are increasingly recognized as an essential asset in the streamlining of modeling efforts. Recently, version 6.0 of ORCA ([Neese et al., 2020](#); [Neese, 2022](#); [Tetenberg et al., 2026](#)) (free for academic use) started offering compound job functionality, which enables the chained execution of some core ORCA modules. However, the code is not open-source and functionality is limited to what is included in the ORCA program (*i.e.* no MLIPs, no similarity

40 pruning, no ensemble processing...). While ORCA workflows are more extensively implemented
41 in WEASEL ([Faccts](#)), this is commercial, closed-source software.

42 Related software to the one presented here include AQME, ([Alegre-Requena et al., 2023](#)) which
43 has what is perhaps the most similar design philosophy. Key differences with FIRECODE, besides
44 code architecture, lie in: 1) the main interaction platform of AQME happening via Jupyter
45 notebooks, while the software of this work favors the use of plain-text input files, keeping the
46 inputs as minimal as possible. 2) The present software having a reactivity-centered set of
47 features, chiefly via fast semiempirical and ML calculators, over AQME's focus on DFT and on
48 the generation of ensemble properties like spectra and molecular descriptors.

49 Other workflow software emerging from the material chemistry community includes atomate2
50 ([A. Ganose et al., 2025](#); [A. M. Ganose et al., 2025](#)), AiIDA ([Huber et al., 2020](#); [Uhrin et al., 2021](#)) and QuAcc ([Rosen, 2026](#)).

52 Reactivity-centered software, primarily for the automated location of transition states, is also
53 gaining popularity. Examples include autodE ([Young et al., 2021](#)) and ML-FSM ([Marks & Gomes, 2025](#)). The latter is indeed interfaced as a utility in FIRECODE, given its light weight
54 and flexibility in working with different ASE calculators.

56 Other examples of related software, specifically targeting the exposure of high-level primitives
57 over structured workflows, include Cuby ([Řezáč, 2016](#)) (MD and high-level DFT) written in
58 Ruby, and the popular ASE ([Bahn & Jacobsen, 2002](#); [Larsen et al., 2017](#)), which is extensively
59 used as a lower-level interface for interacting with calculators in FIRECODE.

60 The development of this software from scratch was motivated by several factors. First, a modern
61 and flexible hub for running computational chemistry workflows, particularly tailored to modeling
62 chemical reactivity and MLIPs, was lacking. In this context, we refer to “reactivity workflows”
63 as the sequence of sequential steps from a three-dimensional molecular representation to a
64 conformationally comprehensive characterization of bond breaking and forming events. This
65 requires access to conformational search modules, efficient multistage ensemble optimization
66 routines, similarity pruning, the ability to use arbitrary geometrical constraints throughout, the
67 ability to optimize to local minima and saddle points alike, and the ability to carry out vibrational
68 analysis and calculate related thermochemical corrections to access enthalpy/entropy/free
69 energy values.

70 While the benefit of developing standalone applications for specific tasks remains, their
71 integration in a single hub is uniquely enabling for automation, and contributes to the
72 popularization of robust atomistic modeling. Second, modern machine-learned interatomic
73 potentials (MLIPs) are often centered around Python libraries like torch ([Ansel et al., 2024](#)),
74 benefitting greatly from a Python-based workflow manager. Additional features of note of this
75 software also include an implicit delta solvation scheme for gas phase-trained MLIPs ¹, and
76 numerical quasi-RRHO thermochemistry ([Grimme, 2012](#)).

77 Software design

78 The software is developed with the philosophy of maintaining the greatest amount of
79 functionality and module interoperability with the minimal list of dependencies, which should
80 ideally be as modern and lean as possible. All core dependencies are distributed via pypi,
81 keeping the installation of the minimal version of the program down to a single command (uv
82 pip install firecode). Non-essential dependencies can be installed based on the desired
83 calculator and interfaces to be used, all via conda.

¹That is, a calculator providing the energy and gradient differences between an atomic structure in an implicit solvent and the same structure in vacuum. These differential contributions are then added to the energy and gradient components of calculations that have no native implementation of solvation models (i.e. gas phase-trained MLIPs).

84 The software implements various calculators (xTB (Bannwarth et al., 2019), tblite (Katbashev
85 et al., 2025), AIMNET2 (Anstine et al., 2025), UMA (Wood et al., 2025)) via ASE (Bahn &
86 Jacobsen, 2002; Larsen et al., 2017), which can be used interchangeably in various ensemble
87 routines. Other core libraries leveraged throughout the codebase are numpy (Harris et al.,
88 2020) for algebraic manipulations, networkx (Hagberg et al., 2008) for graph utilities and
89 prism_pruner (GitHub - Ntampellini/Prism_pruner) for the removal of duplicates. Sella
90 (Hermes et al., 2019, 2021, 2022) provides a saddle-point optimizer, and rdkit (Greg Landrum,
91 2026) is used for ETKDG (Riniker & Landrum, 2015) conformational searches and SMARTS
92 substructure matching. The ML-FSM library (Marks & Gomes, 2025) provides a two-ended
93 TS search utility complementing ASE's implementation of the NEB method (Ásgeirsson et al.,
94 2021).

95 Other standalone conformational search software is instead interfaced via subprocess calls
96 and file-based I/O, like CREST (Pracht et al., 2020, 2024), ORCA's GOAT (Souza, 2025) and
97 racerts (Schmid et al., 2026).

98 The modular nature of the calculators and interfaces allows the definition of complex workflows
99 in plain-text files (input.txt):

```
firecode input.txt -n test_run
```

100 A second standalone executable exposes the core optimizer routines acting directly on structure
101 files:

```
firecode_opt conformers.xyz --opt --freq --solvent toluene [...]
```

102 Contributions by way of adding new routines (or interfaces) to the codebase is made
103 straightforward by design: individual "operators" (routines) are simply functions reading a
104 molecular .xyz file and returning the filename of the processed ensemble. Run information
105 can be accessed from the Embedder class or via environment variables.

```
def center_operator(filename: str, embedder: Embedder) -> str:
    """Example operator centering the molecule."""

    # the Embedder class stores global information
    embedder.avail_gpus # 1
    embedder.options.solvent # "ch2cl2"
    embedder.options.T # 298.15

    # get a copy of the molecule of interest
    mol = embedder.mols[filename]

    # center coordinates
    mol.coords[0] -= np.mean(mol.coords[0], axis=0)

    # save any data you might need later
    embedder.options.centered = True
    embedder.options.last_operator = "center"

    # write to global log
    embedder.log(f"--> Center operator: centered molecule {mol.basename}")

    # write outfile and return its name
    outfile = f"{mol.basename}_centered.xyz"
    mol.to_xyz(outfile)

    return outfile
```

Research impact statement

Since its earliest versions in 2021 (formerly [TSCoDe](#)), this software proved essential in orchestrating asymmetric catalysis workflows with ensembles up to thousands of structures ([Huth et al., 2024](#); [Marvich et al., 2026](#); [Rozema et al., 2025](#); [Tampellini et al., 2024, 2025](#); [Tampellini, Choi, et al., 2025](#)). These included the generation of large conformational ensembles for flexible catalysts and their non-covalent aggregates (often including specific constraints) and the multilevel optimization and pruning of these ensembles to a smaller number of non-redundant structures. These refined ensembles were then directed to higher-level DFT optimization to achieve quantitative insight into stereoselectivity (*i.e.* a predicted $\Delta\Delta G^\ddagger$ value) and obtain other conformationally-comprehensive properties (*i.e.* flexibility, relative populations, number of interaction topologies).

Adoption of FIRECODE is starting to grow, with pypi reporting ~250 downloads per month at the present time. Part of the core code was exported to the `prism_pruner` library ([GitHub - Ntampellini/Prism_pruner](#)) to facilitate its adoption by the computational chemistry startup Rowan ([Rowan | ML-Powered Molecular Design and Simulation — Rowansci.com](#)) (not affiliated with the author).

AI usage disclosure

Generative AI tools (Claude, Gemini) were used in some cases to generate first drafts of Python functions that were then manually curated line-by-line and interfaced with the rest of the codebase by hand. No generative AI tools were used in the writing of this manuscript.

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References

- Alegre-Requena, J. V., Sowndarya S. V, S., Pérez-Soto, R., Alturaifi, T. M., & Paton, R. S. (2023). AQME: Automated quantum mechanical environments for researchers and educators. *WIREs Computational Molecular Science*, 13(5), e1663. <https://doi.org/10.1002/wcms.1663>
- Ansel, J., Yang, E., He, H., Gimelshein, N., Jain, A., Voznesensky, M., Bao, B., Bell, P., Berard, D., Burovski, E., Chauhan, G., Chourdia, A., Constable, W., Desmaison, A., DeVito, Z., Ellison, E., Feng, W., Gong, J., Gschwind, M., ... Chintala, S. (2024). PyTorch 2: Faster machine learning through dynamic python bytecode transformation and graph compilation. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2* (pp. 929–947). ACM. <https://doi.org/10.1145/3620665.3640366>
- Anstine, D. M., Zubatyuk, R., & Isayev, O. (2025). AIMNet2: A neural network potential to meet your neutral, charged, organic, and elemental-organic needs. *Chemical Science*, 16(23), 10228–10244. <https://doi.org/10.1039/d4sc08572h>
- Ásgeirsson, V., Birgisson, B. O., Bjornsson, R., Becker, U., Neese, F., Riplinger, C., & Jónsson, H. (2021). Nudged elastic band method for molecular reactions using energy-weighted

- 149 springs combined with eigenvector following. *Journal of Chemical Theory and Computation*,
150 17(8), 4929–4945. <https://doi.org/10.1021/acs.jctc.1c00462>
- 151 Bahn, S. R., & Jacobsen, K. W. (2002). An object-oriented scripting interface to a legacy
152 electronic structure code [Article]. *Comput. Sci. Eng.*, 4(3), 56–66. <https://doi.org/10.1109/5992.998641>
153
- 154 Bannwarth, C., Ehlert, S., & Grimme, S. (2019). GFN2-xTB—an accurate and broadly
155 parametrized self-consistent tight-binding quantum chemical method with multipole
156 electrostatics and density-dependent dispersion contributions. *Journal of Chemical Theory
157 and Computation*, 15(3), 1652–1671. <https://doi.org/10.1021/acs.jctc.8b01176>
- 158 Faccts. *Weasel - FACCTs* — *faccts.de*. <https://www.faccts.de/weasel/>.
- 159 Ganose, A. M., Sahasrabudhe, H., Asta, M., Beck, K., Biswas, T., Bonkowski, A., Bustamante,
160 J., Chen, X., Chiang, Y., Chrzan, D. C., Clary, J., Cohen, O. A., Ertural, C., Gallant, M.
161 C., George, J., Gerits, S., Goodall, R. E. A., Guha, R. D., Hautier, G., ... Jain, A. (2025).
162 Atomate2: Modular workflows for materials science. *Digital Discovery*, 4(7), 1944–1973.
163 <https://doi.org/10.1039/d5dd00019j>
- 164 Ganose, A., Riebesell, J., George, J., Shen, J., Rosen, A. S., Naik, A. A., nwinner, Wen,
165 M., rdguha1995, Kuner, M., Kaplan, A., Petretto, G., Ertural, C., Horton, M., Zhu,
166 Z., Sahasrabudhe, H., Schmidt, J., Kingsbury, R., Purcell, T., ... Wang, S. (2025).
167 *Materialsproject/atomate2: v0.0.21* (Version v0.0.21). Zenodo. [https://doi.org/10.5281/
168 zenodo.15603088](https://doi.org/10.5281/zenodo.15603088)
- 169 *GitHub - ntampellini/prism_pruner: Similarity pruning engine for conformational ensembles* —
170 *github.com*. https://github.com/ntampellini/prism_pruner.
- 171 Greg Landrum, B. K., Paolo Tosco. (2026). *Rdkit/rdkit: 2026_03_3* (Q1 2026). <https://doi.org/10.5281/zenodo.591637z>
172
- 173 Grimme, S. (2012). Supramolecular binding thermodynamics by dispersion-corrected density
174 functional theory. *Chemistry - A European Journal*, 18(32), 9955–9964. [https://doi.org/
175 10.1002/chem.201200497](https://doi.org/10.1002/chem.201200497)
- 176 Hagberg, A. A., Schult, D. A., & Swart, P. J. (2008). Exploring network structure, dynamics,
177 and function using NetworkX. *Proceedings of the 7th Python in Science Conference*, 11–15.
178 <https://doi.org/10.25080/tcww9851>
- 179 Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D.,
180 Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk,
181 M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., ... Oliphant,
182 T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
183
- 184 Hermes, E. D., Sargsyan, K., Najm, H. N., & Zádor, J. (2019). Accelerated saddle point
185 refinement through full exploitation of partial Hessian diagonalization. *Journal of Chemical
186 Theory and Computation*, 15(11), 6536–6549. <https://doi.org/10.1021/acs.jctc.9b00869>
- 187 Hermes, E. D., Sargsyan, K., Najm, H. N., & Zádor, J. (2021). Geometry optimization speedup
188 through a geodesic approach to internal coordinates. *The Journal of Chemical Physics*,
189 155(9). <https://doi.org/10.1063/5.0060146>
- 190 Hermes, E. D., Sargsyan, K., Najm, H. N., & Zádor, J. (2022). Sella, an open-source
191 automation-friendly molecular saddle point optimizer. *Journal of Chemical Theory and
192 Computation*, 18(11), 6974–6988. <https://doi.org/10.1021/acs.jctc.2c00395>
- 193 Huber, S. P., Zoupanos, S., Uhrin, M., Talirz, L., Kahle, L., Häuselmann, R., Gresch, D.,
194 Müller, T., Yakutovich, A. V., Andersen, C. W., Ramirez, F. F., Adorf, C. S., Gargiulo,
195 F., Kumbhar, S., Passaro, E., Johnston, C., Merkys, A., Cepellotti, A., Mounet, N.,
196 ... Pizzi, G. (2020). AiiDA 1.0, a scalable computational infrastructure for automated

- reproducible workflows and data provenance [Journal Article]. *Scientific Data*, 7(1), 300. <https://doi.org/10.1038/s41597-020-00638-4>
- Huth, S. E., Tampellini, N., Guerrero, M. D., & Miller, S. J. (2024). Catalytic enantioselective sulfoxidation of functionalized thioethers mediated by aspartic acid-containing peptides. *Organic Letters*, 26(32), 6872–6877. <https://doi.org/10.1021/acs.orglett.4c02452>
- Katbashev, A., Stahn, M., Rose, T., Alizadeh, V., Friede, M., Plett, C., Steinbach, P., & Ehlert, S. (2025). Overview on building blocks and applications of efficient and robust extended tight binding. *The Journal of Physical Chemistry A*, 129(10), 2667–2682. <https://doi.org/10.1021/acs.jpca.4c08263>
- Larsen, A. H., Mortensen, J. J., Blomqvist, J., Castelli, I. E., Christensen, R., Duřak, M., Friis, J., Groves, M. N., Hammer, B., Hargus, C., Hermes, E. D., Jennings, P. C., Jensen, P. B., Kermode, J., Kitchin, J. R., Kolsbjerg, E. L., Kubal, J., Kaasbjerg, K., Lysgaard, S., ... Jacobsen, K. W. (2017). The atomic simulation environment—a Python library for working with atoms. *Journal of Physics: Condensed Matter*, 29(27), 273002. <http://stacks.iop.org/0953-8984/29/i=27/a=273002>
- Marks, J., & Gomes, J. (2025). Incorporation of internal coordinates interpolation into the freezing string method. *Journal of Chemical Theory and Computation*, 21(23), 12110–12120. <https://doi.org/10.1021/acs.jctc.5c01492>
- Marvich, H. M., Langner, O. C., Tampellini, N., & Miller, S. J. (2026). A bifunctional aminoxyl-bipyridine peptide catalyst for the atroposelective copper-catalyzed aerobic oxidation of biaryl diols. *Journal of the American Chemical Society*, 148(22), 23032–23041. <https://doi.org/10.1021/jacs.6c04560>
- Neese, F. (2022). Software update: The ORCA program system—version 5.0. *WIREs Computational Molecular Science*, 12(5), e1606. <https://doi.org/10.1002/wcms.1606>
- Neese, F., Wennmohs, F., Becker, U., & Riplinger, C. (2020). The ORCA quantum chemistry program package. *The Journal of Chemical Physics*, 152(22), 224108. <https://doi.org/10.1063/5.0004608>
- Pracht, P., Bohle, F., & Grimme, S. (2020). Automated exploration of the low-energy chemical space with fast quantum chemical methods. *Physical Chemistry Chemical Physics*, 22(14), 7169–7192. <https://doi.org/10.1039/C9CP06869D>
- Pracht, P., Grimme, S., Bannwarth, C., Bohle, F., Ehlert, S., Feldmann, G., Gorges, J., Müller, M., Neudecker, T., Plett, C., Spicher, S., Steinbach, P., Wesołowski, P. A., & Zeller, F. (2024). CREST—a program for the exploration of low-energy molecular chemical space. *The Journal of Chemical Physics*, 160(11), 114110. <https://doi.org/10.1063/5.0197592>
- Řezáč, J. (2016). Cuby: An integrative framework for computational chemistry. *Journal of Computational Chemistry*, 37(13), 1230–1237. <https://doi.org/10.1002/jcc.24312>
- Riniker, S., & Landrum, G. A. (2015). Better informed distance geometry: Using what we know to improve conformation generation. *Journal of Chemical Information and Modeling*, 55(12), 2562–2574. <https://doi.org/10.1021/acs.jcim.5b00654>
- Rosen, A. (2026). *Quacc – the quantum accelerator*. Zenodo. <https://doi.org/10.5281/ZENODO.7720998>
- Rowan | ML-Powered Molecular Design and Simulation — rowansci.com. <https://rowansci.com/>.
- Rozema, S. D., Tampellini, N., Rein, J., Sigman, M. S., Lin, S., & Miller, S. J. (2025). Experimental lineage and computational analysis of a general aminoxyl-based oxidation catalyst: Generality from substrate-specific interactions. *ACS Catalysis*, 15(20), 17548–17557. <https://doi.org/10.1021/acscatal.5c05893>

- 244 Schmid, S. P., Seng, H., Kläy, T., & Jorner, K. (2026). Rapid generation of transition-state
245 conformer ensembles via constrained distance geometry. *Journal of Chemical Information*
246 *and Modeling*, 66(5), 2777–2790. <https://doi.org/10.1021/acs.jcim.5c02794>
- 247 Souza, B. de. (2025). GOAT: A global optimization algorithm for molecules and atomic
248 clusters. *Angewandte Chemie International Edition*, 64(18), e202500393. <https://doi.org/10.1002/anie.202500393>
- 249
- 250 Tampellini, N., Choi, E. S., & Miller, S. J. (2025). Peptide-catalyzed asymmetric amination
251 of sulfenamides enabled by DFT-guided catalyst optimization. *Journal of the American*
252 *Chemical Society*, 147(44), 41122–41129. <https://doi.org/10.1021/jacs.5c15618>
- 253 Tampellini, N., Mercado, B. Q., & Miller, S. J. (2024). Scaffold-oriented asymmetric catalysis:
254 Conformational modulation of transition state multivalency during a catalyst-controlled
255 assembly of a pharmaceutically relevant atropisomer. *Chemistry – A European Journal*,
256 30(30), e202401109. <https://doi.org/10.1002/chem.202401109>
- 257 Tampellini, N., Mercado, B. Q., & Miller, S. J. (2025). Enantiocontrolled cyclization to
258 form chiral 7- and 8-membered rings unified by the same catalyst operating with different
259 mechanisms. *Journal of the American Chemical Society*, 147(5), 4624–4630. <https://doi.org/10.1021/jacs.4c17080>
- 260
- 261 Tetenberg, T., Neugebauer, H., Plett, C., Santhosh, N., Bursch, M., & Riplinger, C. (2026).
262 ORCA meets python - the ORCA python interface OPI. *Journal of Chemical Theory and*
263 *Computation*, 22(10), 4951–4967. <https://doi.org/10.1021/acs.jctc.5c02141>
- 264 Uhrin, M., Huber, S. P., Yu, J., Marzari, N., & Pizzi, G. (2021). Workflows in AiiDA:
265 Engineering a high-throughput, event-based engine for robust and modular computational
266 workflows. *Computational Materials Science*, 187, 110086. [https://doi.org/10.1016/j.](https://doi.org/10.1016/j.commatsci.2020.110086)
267 [commatsci.2020.110086](https://doi.org/10.1016/j.commatsci.2020.110086)
- 268 Wood, B. M., Dzamba, M., Fu, X., Gao, M., Shuaibi, M., Barroso-Luque, L., Abdelmaqsoud,
269 K., Gharakhanyan, V., Kitchin, J. R., & Levine, D. S. (2025). UMA: A family of universal
270 models for atoms. *arXiv Preprint*. <https://doi.org/10.48550/arXiv.2506.23971>
- 271 Young, T. A., Silcock, J. J., Sterling, A. J., & Duarte, F. (2021). autodE: Automated
272 calculation of reaction energy profiles – application to organic and organometallic reactions.
273 *Angewandte Chemie International Edition*, 60(8), 4266–4274. [https://doi.org/10.1002/](https://doi.org/10.1002/anie.202011941)
274 [anie.202011941](https://doi.org/10.1002/anie.202011941)